

Worksheet W5: The Power Rule

"Bring the exponent down in front, then subtract 1 from the exponent."

Chapter 5 · 15 Problems · Name: _____ Date: _____

Power Rule: If $f(x) = x^n$, then $f'(x) = n \times x^{(n-1)}$

Problem 1

Find the derivative of $f(x) = x^2$.

💡 *Hint: Bring 2 down in front, subtract 1 from the exponent.*

Problem 2

Find the derivative of $f(x) = x^3$.

💡 *Hint: Bring 3 down in front, subtract 1 from the exponent.*

Problem 3

Find the derivative of $f(x) = x^4$.

💡 *Hint: Bring 4 down in front, subtract 1 from the exponent.*

Problem 4

Find the derivative of $f(x) = x^7$.

💡 *Hint: Bring 7 down in front, subtract 1 from the exponent.*

Problem 5

Find the derivative of $f(x) = 2x^3$.

💡 Hint: The coefficient stays in front. Multiply the coefficient by the exponent.

Problem 6

Find the derivative of $f(x) = 3x^2$.

💡 Hint: Coefficient $\times$ exponent, then reduce exponent by 1.

Problem 7

Find the derivative of $f(x) = 4x^3$.

💡 Hint: $4 \times 3 = 12$, exponent $3 \rightarrow 2$.

Problem 8

Find the derivative of $f(x) = 5x^4$.

💡 Hint: $5 \times 4 = 20$, exponent $4 \rightarrow 3$.

Problem 9

Find the derivative of $f(x) = x^5 + x^2$.

💡 Hint: Find the derivative of each term separately, then add them.

Problem 10

Find the derivative of $f(x) = x^2 + x^3 + x^4$.

💡 Hint: Derivative of x^2 is $2x$, of x^3 is $3x^2$, of x^4 is $4x^3$.

Problem 11

Find the derivative of $f(x) = 2x^5 + 3x^3$.

💡 Hint: Derivative of $2x^5$ is $10x^4$. Derivative of $3x^3$ is $9x^2$.

Problem 12

Find the derivative of $f(x) = 4x^2 + 5$.

💡 Hint: The derivative of a constant is 0.

Problem 13

Find the derivative of $f(x) = 3x^6 + 2x^4 + x$.

💡 Hint: The derivative of x is 1 (since $x = x^1$).

Problem 14

Find the derivative of $f(x) = 2x^4 + 5x^2 + 3$.

💡 Hint: The derivative of a constant is 0.

Problem 15

Find the derivative of $f(x) = x^6 + 4x^3 + 2x + 7$.

💡 *Hint: Derivative of x^6 is $6x^5$, of $4x^3$ is $12x^2$, of $2x$ is 2 , of 7 is 0 .*